

Geospatial Analysis in Forensic GIS

Detailed course syllabus structured in the style of a university semester syllabus

Course:	Geospatial Analysis in Forensic GIS	Level:	Graduate / advanced undergraduate
Format:	Lecture + lab / applied session	Credits:	3
Instructor:	Farnoosh Roozkhosh	Delivery:	In-person, hybrid, or online adaptable

Course Content

This course introduces concepts, theories, and techniques for geospatial analysis in forensic GIS. It examines how spatial data, GIS methods, and spatial statistical reasoning can support forensic interpretation, investigation, and defensible decision-making. The course covers foundational and advanced topics including spatial data uncertainty, exploratory spatial data analysis, point pattern analysis, polygon-based analysis, spatial autocorrelation, hotspot analysis, spatial regression, geographically weighted regression, spatiotemporal analysis, and network and movement analysis.

The course also emphasizes model validation, forensic defensibility, ethics, privacy, and legal responsibility when using spatial evidence. Along with the lectures, regular labs and applied exercises strengthen analytical understanding and build practical GIS skills.

Course Objectives

- Develop an understanding of core concepts in forensic GIS and geospatial analysis.
- Become familiar with spatial reasoning, spatial dependence, spatial heterogeneity, and uncertainty.
- Apply GIS and spatial analytical methods to forensic and public-safety related questions.
- Interpret point, polygon, network, and spatiotemporal spatial patterns.
- Evaluate when global statistical models are insufficient for spatial data.
- Apply and interpret local spatial models such as Geographically Weighted Regression (GWR).
- Communicate geospatial results in plain language for analytical and professional audiences.
- Critically assess model validity, defensibility, limitations, ethics, and privacy concerns.

Prerequisite

Introductory GIS or equivalent experience with maps, spatial data, and basic GIS operations is recommended. Introductory statistics is helpful but not required.

Provided Texts

No single required textbook is necessary. Readings will be selected from lecture slides, instructor notes, journal articles, software documentation, and short tutorials. Representative materials may include resources in forensic GIS, GIScience, spatial analysis, public health GIS, crime analysis, and ethics of spatial evidence.

- Lecture slides and instructor notes
- Selected journal articles in forensic GIS, GIScience, spatial epidemiology, public health GIS, and spatial statistics
- ArcGIS Pro and GeoDa tutorials and documentation

- Selected readings on ethics, privacy, and defensibility in GIS-based analysis

Course Work

Five types of tasks are designed to help students achieve the course objectives and help evaluate student performance.

Item	Description
1. Pop Quizzes / Checks	Short quizzes or reflection checks may be given throughout the semester to reinforce weekly concepts and readings. The lowest score may be dropped at the end of the semester.
2. Article Review	Each student reviews one recent journal article related to forensic GIS, spatial analysis, or a related applied topic and gives a short 5–8 minute presentation.
3. Applied Project	Graduate students complete an individual project and 15–20 page term paper. Undergraduate students complete a guided or self-selected project approved by the instructor and submit a 5–10 page report.
4. Discussion / Debate	Students participate in one structured discussion or debate on a controversial issue related to forensic mapping, spatial evidence, privacy, or ethical interpretation.
5. Labs	Regular labs provide hands-on experience with GIS and spatial analytical methods using ArcGIS Pro, GeoDa, and other supporting tools as needed.

Grading Policy

Course Work	Weight
Weekly quizzes / short checks	15%
Article review	10%
Discussion / debate	10%
Project	30%
Labs	35%

Other Items

Due Dates. All assignments must be completed on time. Late submissions may be accepted with a penalty of 10% of the assignment grade per day late, unless prior arrangements are made or there is documentation of a medical or family emergency.

Assignment Format. Written assignments should be typed, clearly organized, and submitted in a professional academic format unless otherwise noted.

Academic Honesty. All academic work must be original. Plagiarism, fabrication, or submitting others' work as one's own will be treated as academic misconduct.

Use of Generative AI. Students may use generative AI tools for brainstorming, organization, or exploratory support, but may not submit AI-generated text, code, or analysis as their own work without proper disclosure and citation. Students are responsible for checking all outputs for errors, bias, and unsupported claims.

Make-up Work. Make-up work is granted only in documented cases of medical, family, or university-approved emergencies.

Topics and Course Schedule

TENTATIVE PLAN, SUBJECT TO FINE-TUNING

Week	Lecture	Reading / Preparation	Lab / Applied Activity
1	Course introduction; what is forensic GIS?	Intro lecture notes	No lab / software setup
2	Introduction to forensic GIS and geospatial modeling	Lecture notes + selected article	GIS data familiarization
3	Foundations of statistical reasoning for spatial analysis	Lecture notes	Descriptive statistics and mapping exercise
4	Spatial data types, scale, and uncertainty	Lecture notes	Data quality and scale exercise
5	Exploratory spatial data analysis in forensic GIS	Lecture notes	Thematic mapping and exploratory analysis
6	Point pattern analysis in forensic GIS	Selected reading	Point event mapping / density / pattern exercise
7	Polygon-based analysis in forensic GIS	Selected reading	Rate mapping / choropleth interpretation
8	Spatial autocorrelation in forensic GIS	Lecture notes	Moran's I and spatial autocorrelation report
9	Hotspot and cluster analysis	Selected reading	Hotspot analysis lab
10	Spatial regression in forensic GIS	Lecture notes	OLS setup and residual interpretation
11	Geographically Weighted Regression (GWR) in forensic GIS	GWR lecture notes / module slides	GWR workflow and local coefficient interpretation

Week	Lecture	Reading / Preparation	Lab / Applied Activity
12	Spatiotemporal analysis in forensic GIS	Selected reading	Time-enabled mapping / temporal comparison
13	Network and movement analysis in forensic GIS	Selected reading	Route / service area / movement analysis
14	Model validation and forensic defensibility	Lecture notes	Residuals, limitations, defensibility exercise
15	Ethics, privacy, and legal issues in forensic GIS	Selected reading	Case discussion / final project workshop

Highlighted Session in This Course

Geographically Weighted Regression (GWR) in Forensic GIS. One session of this course focuses on GWR as a local statistical method for analyzing spatially varying relationships. In this session, students move from identifying clustered patterns to asking whether relationships between overdose mortality and socioeconomic factors remain constant across counties or vary by place. The lesson introduces spatial heterogeneity, contrasts stationary and non-stationary relationships, explains why OLS may be insufficient, and uses Moran's I and GWR results to support place-specific interpretation.

Potential Final Project Topics

- Overdose mortality and social vulnerability
- Crime clustering and neighborhood context
- Service accessibility and emergency response
- Environmental exposure and spatial inequity
- Route, movement, or activity-space analysis
- Spatial patterns in public safety or forensic investigation

Suggested Journal Areas

Students are encouraged to consult journals in GIScience, spatial analysis, criminology, public health, and forensic applications.

• International Journal of Geographical Information Science	• Health & Place
• Transactions in GIS	• Social Science & Medicine
• Cartography and Geographic Information Science	• Crime Science
• Applied Geography	• International Journal of Health Geographics
• Computers, Environment and Urban Systems	